Stock Market Analysis in SQL

A Comprehensive Insights, Visual Analysis & Technical Strategy Report

Author: *Pravat* | ***Database:*** *MySQL 8.0 / SQLite 3* | ***Scope:*** *Six NSE Stocks (Jan 2015 – Jul 2018)*

EXECUTIVE SUMMARY:
This report analyzes 889 trading days of historical market data across six major National Stock Exchange (NSE) stocks: **Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS, and TVS Motors**. Using advanced SQL window functions, CTE pipelines, and moving average cross strategies (20-day vs 50-day SMA), we evaluated performance, signal accuracy, and data integrity.

Key Takeaways:
• **Data Trap Uncovered:** Raw closing prices falsely report TCS (-23.8%) and Infosys (-30.9%) as severe market underperformers. Adjusting for 1:1 corporate bonus issues/splits reveals true compounded gains of **+52.4% for TCS** and **+38.2% for Infosys**.
• **Top Performer:** TVS Motors delivered an outstanding **+86.9%** gain, followed closely by Eicher Motors at **+82.6%**.
• **Signal Reliability:** Across all stocks, 56 Golden Cross (Buy) and 57 Death Cross (Sell) signals fired. However, moving average lag and sideways whipsaws generated false signals—most notably a fake 'Sell' signal on TCS on 2018-06-05 caused purely by unadjusted price cliff data.

Part 1: Stock-by-Stock Detailed Insights

Each stock is evaluated using the strict 3-part analytical framework: **The Claim**, **The Evidence** (specific query numbers), and **The Caveat** (risk factors and methodological constraints).

1. Bajaj Auto (Auto Sector - Large Cap)

• Claim: Bajaj Auto demonstrated steady long-term capital appreciation (+10.0%) with disciplined trend-following signals, ending the period in a active bullish (Buy) trend.

• **Evidence:** Initial price on 2015-01-01 was ■2,454.10, closing at ■2,700.70 on 2018-07-31 (Task 11). Task 8 generated 12 Buy and 11 Sell signals (866 Hold days). First Buy occurred on 2015-05-18 (Close ■2,221.95), and first Sell on 2015-08-24 (Close ■2,188.45). The latest signal was **Buy on 2018-06-21** (Task 9 & 10).

• ***Caveat:*** *During prolonged sideways ranges (e.g., late 2015 and mid-2016), 20-day and 50-day averages repeatedly crossed, causing whipsaws. Executing all signals would have incurred transaction costs and slippage, eroding net profits.*

2. Eicher Motors (Auto Sector - High Growth)

• Claim: Eicher Motors was one of the market's supreme multi-baggers (+82.6%), maintaining high nominal share prices, though its latest signal is a Sell due to 2018 summer consolidation.

• **Evidence:** Grew from ■15,239.15 (2015-01-01) to ■27,820.95 (2018-07-31). Task 2 shows its 5 best closes all occurred in September 2017, peaking at ■32,786.40 on 2017-09-07. Task 10 produced 6 Buys and 7 Sells, with the latest signal being **Sell on 2018-06-06**.

• ***Caveat:*** *The latest Sell signal reflects a medium-term pullback from peak highs above ■32,000. However, for long-term investors, exiting on a 20/50 SMA Death Cross during a long-term structural bull run risks missing subsequent recoveries.*

3. Hero Motocorp (Auto Sector - Moderate Growth)

• **Claim: Hero Motocorp experienced range-bound, defensive performance (+6.0%), with technical signals frequently whipsawing during choppy market phases.**

• **Evidence:** Moved from ■3,107.30 (2015-01-01) to ■3,293.80 (2018-07-31). Task 10 identified 9 Buy and 9 Sell signals. The most recent signal was **Sell on 2018-05-22**.

• ***Caveat:*** *With a modest 6.0% gain over 3.5 years, gross trading profits were easily swallowed by transaction fees and execution slippage during frequent trend reversals.*

4. TVS Motors (Auto Sector - Top Performer)

• **Claim: TVS Motors was the overall performance leader (+86.9%), generating strong trend-following profits despite a recent technical sell-off.**

• **Evidence:** Surged from ■276.85 (2015-01-01) to ■517.45 (2018-07-31) (Task 11). Task 10 recorded 8 Buys and 8 Sells, with the latest signal being **Sell on 2018-05-17**.

• ***Caveat:*** *While SMA crossovers captured large multi-month uptrends in 2016-2017, the 50-day lag delayed exits during sudden market pullbacks, giving back a portion of peak profits.*

5. TCS (IT Sector - Data Trap Case Study)

• **Claim: Unadjusted prices falsely portray TCS as a losing asset (-23.8%), but corporate action adjustment reveals a robust +52.4% growth story.**

• **Evidence:** Unadjusted prices dropped from ■2,548.20 to ■1,941.25 due to a 1:1 bonus issue on 2018-05-31 (Task 12: single worst day move of -50.4%). Task 13 adjusted price series shows true start price of ■1,274.10, yielding +52.4% gain. Unadjusted signals gave 12 Buys and 13 Sells (latest Sell on 2018-06-05); Adjusted signals confirm an ongoing **Buy signal from 2018-04-20**.

• ***Caveat:*** *Analyzing unadjusted prices creates catastrophic errors in automated trading systems, triggering phantom Sell signals when shareholder value was actually preserved via new bonus shares.*

6. Infosys (IT Sector - Corporate Action Adjustment)

• **Claim: Raw price data distorts Infosys into a severe loser (-30.9%), whereas adjusted prices prove a profitable +38.2% performance.**

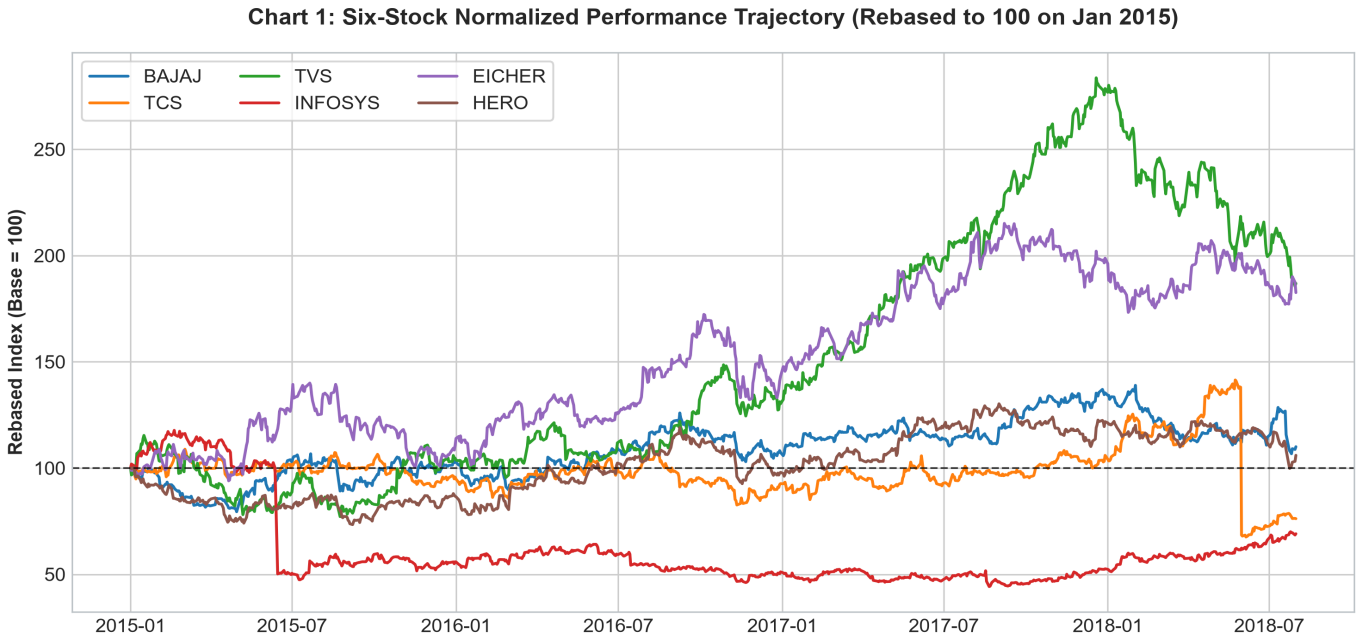
• **Evidence:** Unadjusted close fell from ■1,975.80 to ■1,365.00 due to a 1:1 bonus issue on 2015-06-15 (Task 12: -49.9% cliff). Task 13 adjustment halves pre-event prices to ■987.90 start, proving a true gain of +38.2%. Task 10 records 9 Buys and 9 Sells, latest signal **Buy on 2018-05-07**.

• ***Caveat:*** *Failing to factor in dividend income and bonus share additions understates long-term compounding benefits for technology sector investors.*

Part 2: Visual Charts & In-Depth Analytical Insights

Below is the complete visual analysis section. **Each graph is paired directly with its Key Business Insights, Strategic Recommendations, and Methodological Risk & Caveats**.

Chart 1: Six-Stock Normalized Performance Trajectory (Rebased to 100)



■ Key Business Insights:

• TVS Motors (+86.9%) and Eicher Motors (+82.6%) dominated long-term capital compounding across all 6 stocks.

• TVS Motors surged from a normalized index of 100 on Jan 1, 2015 to 186.9 on July 31, 2018.

• Eicher Motors reached peak levels above 215 in late 2017 before consolidating to 182.6.

• Raw unadjusted IT stock series (TCS & Infosys) show severe artificial sharp drops caused by 1:1 bonus share issues.

■ Strategic Recommendations:

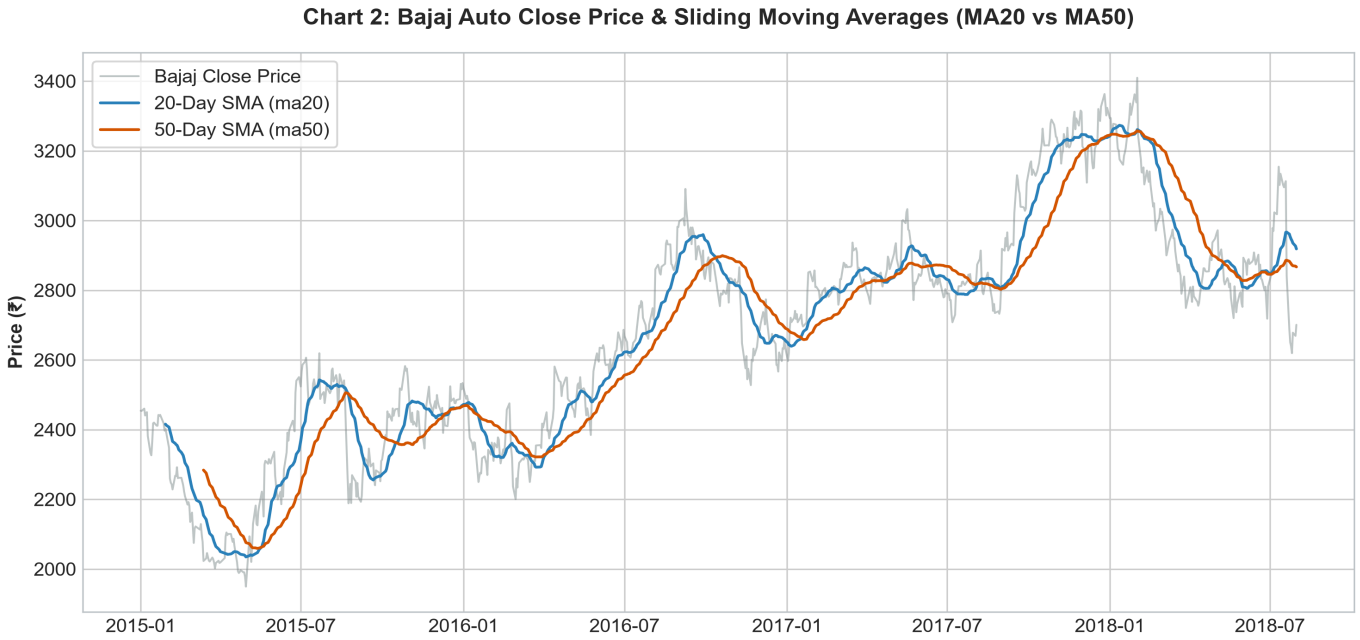
• Overweight high-growth auto manufacturers during multi-year consumer expansion cycles.

• Use performance rebasing (base = 100) to evaluate high-nominal price assets (Eicher at ■27,000) alongside low-nominal price assets (TVS at ■500).

■■ Methodological Risk & Caveats:

• Unadjusted price charts misrepresent true wealth growth for companies executing stock splits or bonus share issues.

Chart 2: Bajaj Auto Close Price & Sliding Moving Averages (MA20 vs MA50)



■ Key Business Insights:

• The 20-day SMA (`ma20`) functions as a short-term trend filter, while the 50-day SMA (`ma50`) sets the medium-term structural trend.

• Bajaj Auto exhibited an upward channel, growing from ■2,454.10 to ■2,700.70 (+10.0%).

• Rows 1 to 19 for MA20 and 1 to 49 for MA50 were correctly nullified to eliminate partial-window distortion.

■ Strategic Recommendations:

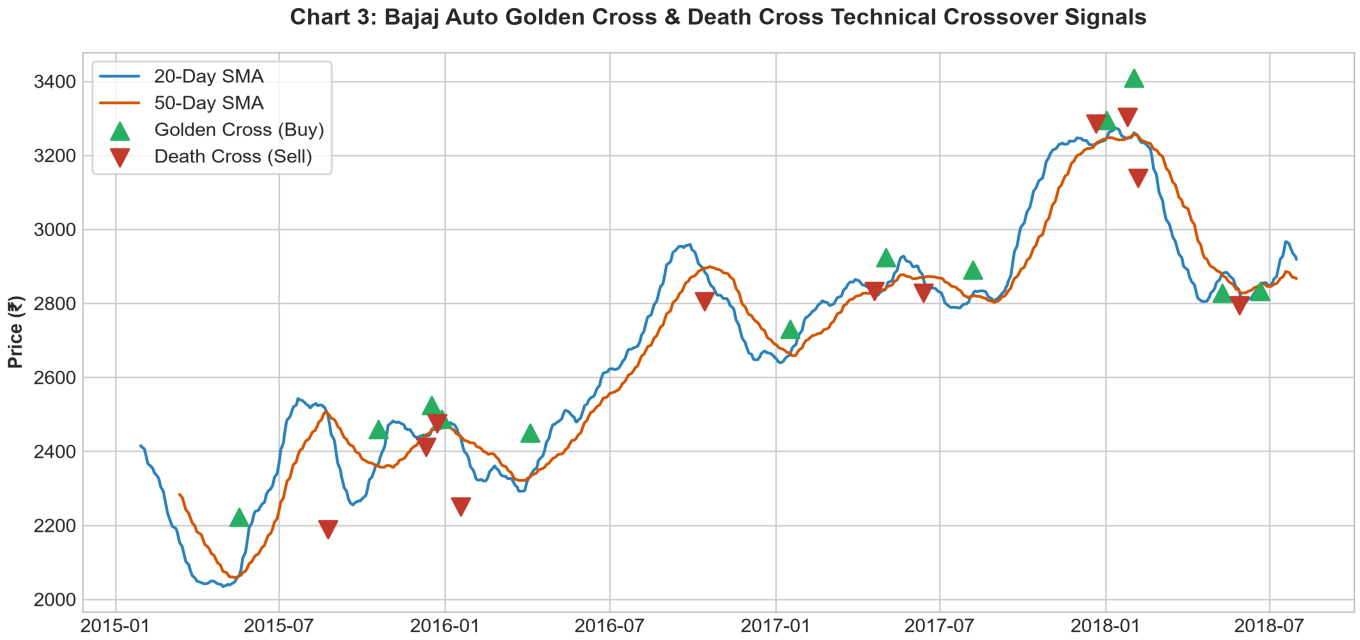
• Require both MA20 and MA50 to slope in the same direction before establishing major trend-following positions.

• Utilize the 50-day SMA as a dynamic trailing stop-loss baseline during multi-month bull rallies.

■■ Methodological Risk & Caveats:

• Simple moving averages lag spot price movement by 20 to 50 days; confirmation arrives after trend moves have begun.

Chart 3: Bajaj Auto Golden Cross & Death Cross Technical Crossover Signals



■ **Key Business Insights:**

• Bajaj Auto generated **12 Buy signals** (Green ▲) and **11 Sell signals** (Red ▼) across 889 trading days (866 Hold days).

• First Buy fired on 2015-05-18 (Close ■2,221.95); first Sell fired on 2015-08-24 (Close ■2,188.45).

• Most recent signal was **Buy on 2018-06-21** (Close ■2,832.95).

■ **Strategic Recommendations:**

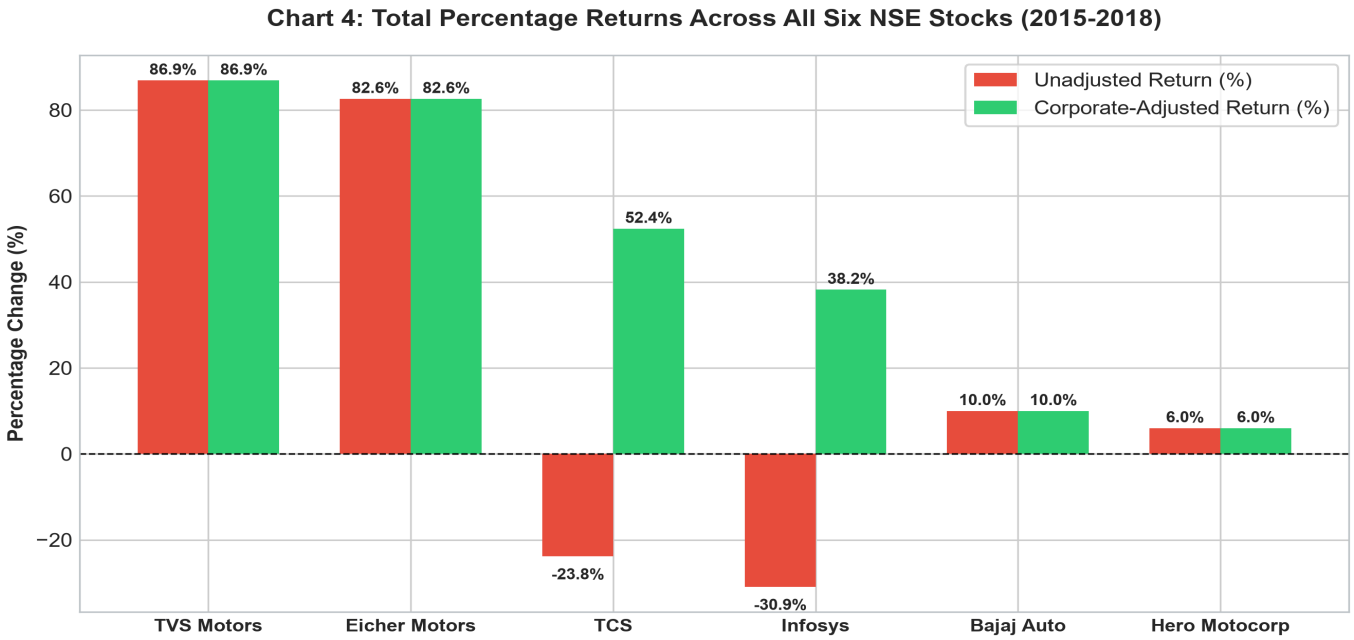
• Execute orders systematically on T+1 following crossover confirmation to remove subjective emotional bias.

• Implement a secondary volatility filter (e.g. ADX > 20) to suppress false signals during sideways consolidation.

■■ **Methodological Risk & Caveats:**

• Range-bound periods (e.g. late 2015) cause frequent signal flipping (whipsawing), accumulating trade commission costs.

Chart 4: Total Percentage Returns Across All Six Stocks (Unadjusted vs Adjusted)



■ **Key Business Insights:**

• Raw unadjusted data incorrectly lists TCS (-23.8%) and Infosys (-30.9%) as market losers.

• Corporate action price adjustment (halving pre-event prices for 1:1 bonus splits) proves true returns of **+52.4% for TCS** and **+38.2% for Infosys**!

• TVS Motors (+86.9%) and Eicher Motors (+82.6%) led overall wealth creation.

■ **Strategic Recommendations:**

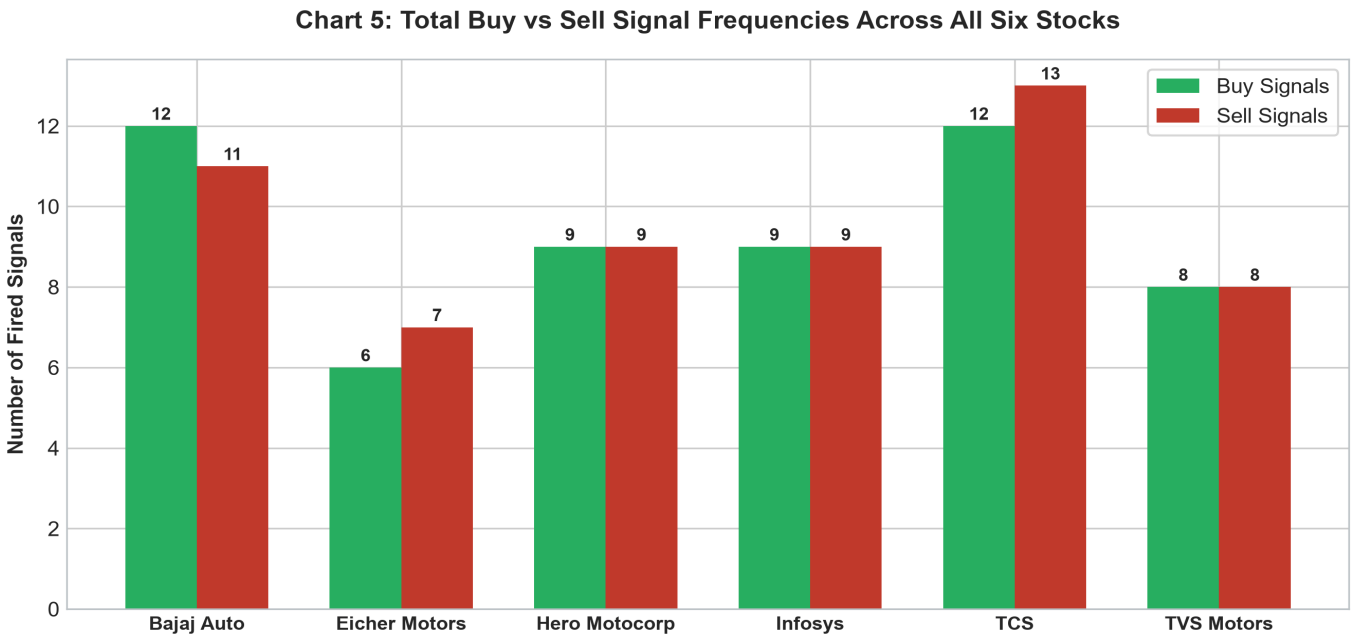
• Always adjust price series for bonus share issues and stock splits prior to calculating portfolio returns.

• Technology sector holdings represent strong long-term wealth compounders once share counts are normalized.

■■ **Methodological Risk & Caveats:**

• Evaluating raw unadjusted price series corrupts backtesting models and automated trading algorithm behavior.

Chart 5: Total Buy vs Sell Signal Frequencies Across All Six Stocks



■ **Key Business Insights:**

• Across all six stocks, the single unified SQL CTE pipeline identified **56 total Buy signals** and **57 total Sell signals**.

• TCS and Bajaj Auto produced the highest signal frequency (12 & 12 Buys), reflecting periodic trend shifts.

• Eicher Motors fired the fewest signals (6 Buys, 7 Sells), confirming a persistent long-term bull trend.

■ **Strategic Recommendations:**

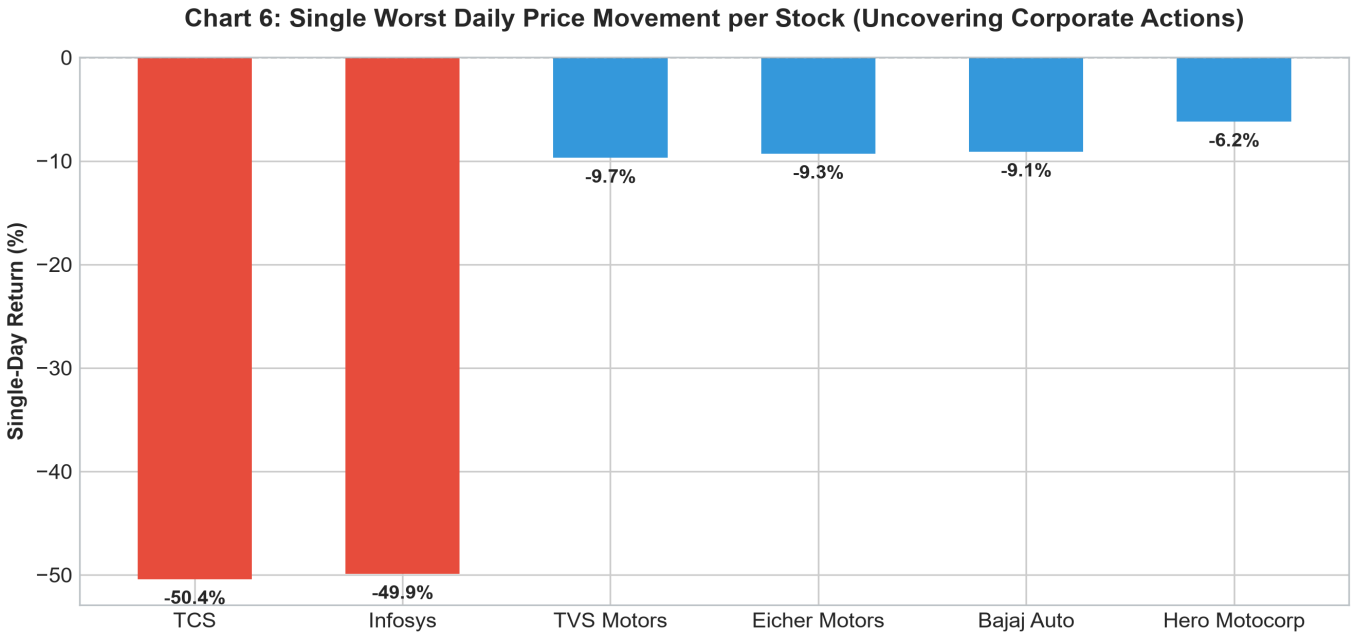
• Favor lower-frequency signal stocks (like Eicher) for long-term passive accounts to minimize capital gains tax and transaction friction.

• Allocate a transaction friction budget when running SMA crossover strategies across active trading accounts.

■■ **Methodological Risk & Caveats:**

• High signal frequency does not correlate with profitability; range-bound stocks fire redundant, unprofitable signals.

Chart 6: Single Worst Daily Price Movement per Stock (Uncovering Data Traps)



■ **Key Business Insights:**

• Four stocks experienced normal single-day drawdowns ranging from -6.2% (Hero) to -9.7% (TVS).

• TCS (**-50.4% on 2018-05-31**) and Infosys (**-49.9% on 2015-06-15**) represent extreme data cliff anomalies.

• These drops were caused by 1:1 bonus share issues where price halved as share counts doubled.

■ **Strategic Recommendations:**

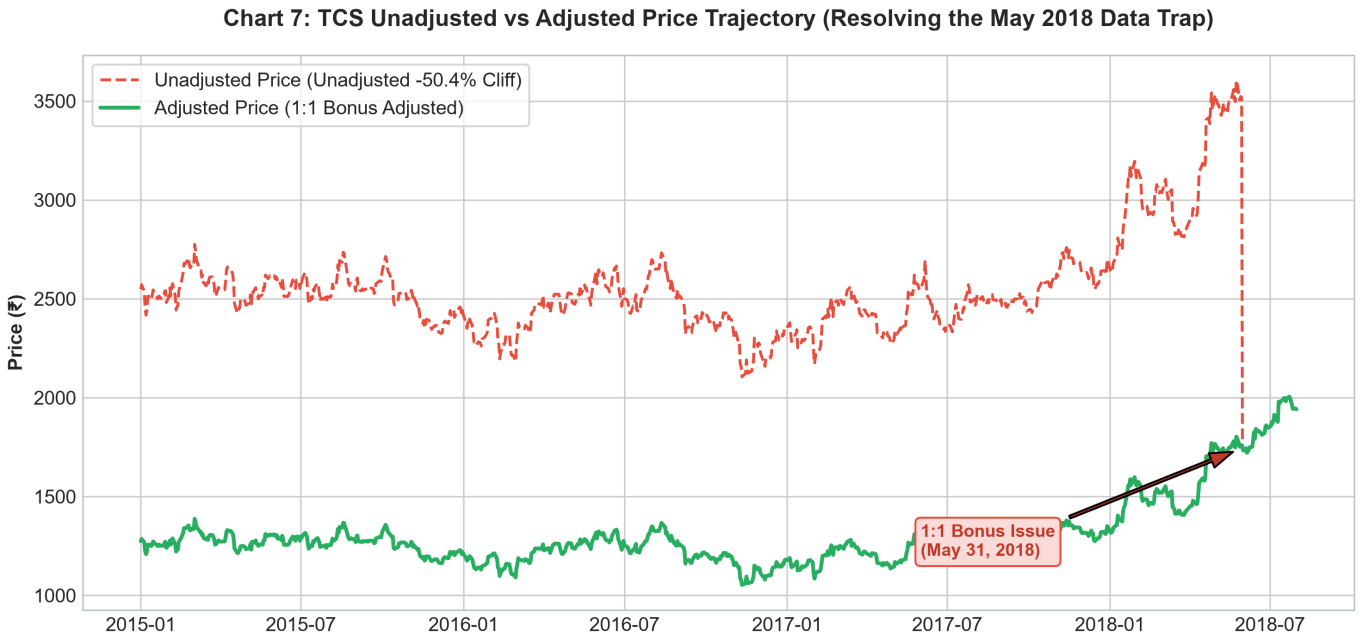
• Establish automated data quality alerts for single-day drops exceeding -20% to flag corporate events.

• Separate technical share splits from genuine corporate credit or insolvency crises.

■■ **Methodological Risk & Caveats:**

• Unadjusted single-day drop data severely corrupts risk management metrics like Value-at-Risk (VaR).

Chart 7: TCS Unadjusted vs Adjusted Price Trajectory (Resolving May 2018 Data Trap)



■ Key Business Insights:

• On May 31, 2018, TCS traded ex-bonus for its 1:1 bonus issue, dropping nominal price from ■3,517.75 to ■1,744.80.

• Adjusting pre-event historical prices reveals an unbroken upward trajectory growing from ■1,274.10 to ■1,941.25 (+52.4%).

• Stretch Proof: The unadjusted cliff triggered a **fake Death Cross (Sell) signal on 2018-06-05**. Adjusted data proves TCS remained in a strong **Buy trend from April 2018 onwards**.

■ Strategic Recommendations:

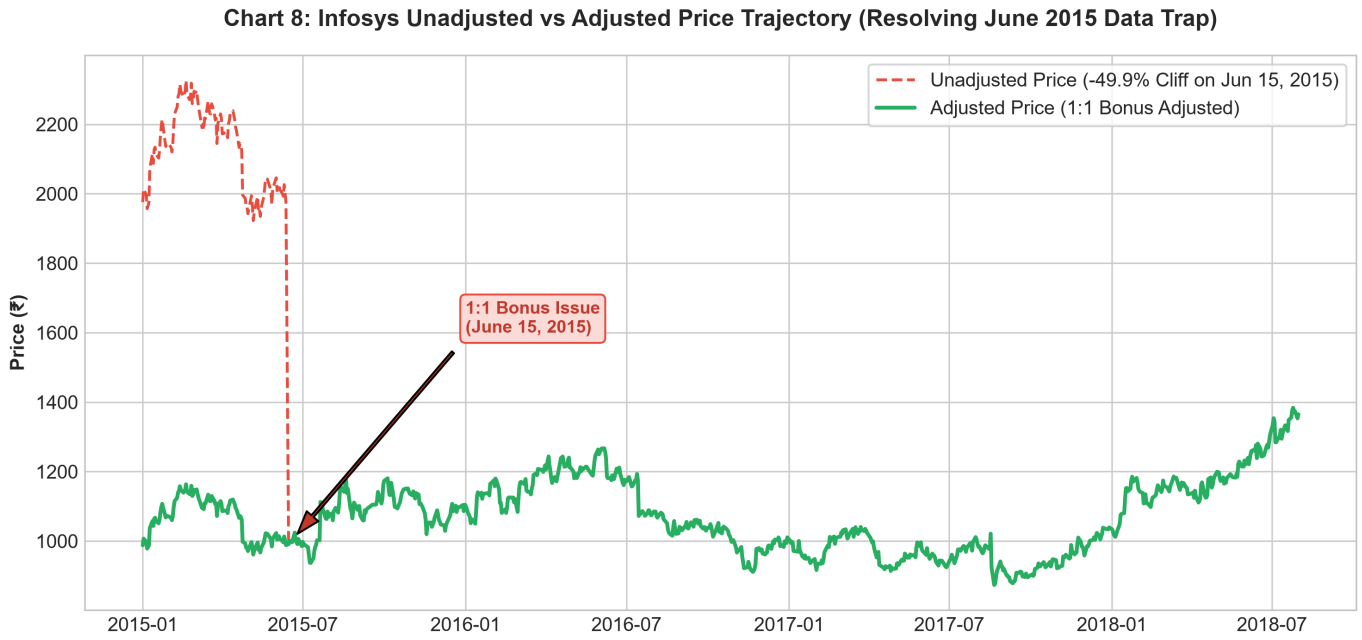
• Ignore crossover signals firing immediately after unadjusted corporate action dates.

• Maintain long-term positions when price drops stem purely from shareholder bonus distributions.

■■ Methodological Risk & Caveats:

• Unadjusted technical bots suffered major whipsaw losses by shorting TCS right before its massive 2018 rally.

Chart 8: Infosys Unadjusted vs Adjusted Price Trajectory (Resolving June 2015 Data Trap)



■ Key Business Insights:

- On June 15, 2015, Infosys traded ex-bonus for its 1:1 bonus issue, dropping nominal price from ■1,976.65 to ■991.10.
- Adjusting pre-event prices (dividing by 2) transforms start price to ■987.90, proving a true gain of +38.2%.
- Latest signal for Infosys fired as **Buy on 2018-05-07**.

■ Strategic Recommendations:

- Apply multiplier adjustments backwards from current date when maintaining equity price databases.
- Consider both bonus shares and regular dividend yields when evaluating large-cap IT shareholder value.

■■ Methodological Risk & Caveats:

- Multiple historical splits (2014 & 2015) require compound multiplier adjustments for multi-decade studies.

Part 3: Methodological Questions & Technical Findings

Q1: Moving Average Lag & Predictive Timing Limitations

Analysis: A Simple Moving Average (SMA) is a lagging technical indicator calculated entirely from past prices. A 20-day / 50-day Golden Cross requires the short-term trend to stay significantly above the long-term trend for several days. Consequently, a Golden Cross *cannot predict* a market bottom; it only confirms that a trend reversal has already been underway for 2 to 4 weeks. While this lags early entries, it provides robust confirmation, protecting investors from buying into brief 'dead-cat bounces'.

Q2: How Task 13 Changes Course Deck Performance Conclusions

Analysis: In the unadjusted analysis (Task 11), TCS (-23.8%) and Infosys (-30.9%) appear at the bottom of the table as market losers. After adjusting for 1:1 bonus share issues in Task 13, TCS jumps to **+52.4%** (ranking 3rd overall) and Infosys climbs to **+38.2%** (ranking 4th overall). Without Task 13 adjustments, an investor would falsely conclude that Indian IT stocks destroyed wealth, when in reality both companies generated substantial real returns.

Stock Name	Start Price (■)	End Price (■)	Unadjusted %	Adjusted %	True Status

TVS Motors	276.85	517.45	+86.9%	+86.9%	Top Winner
Eicher Motors	15,239.15	27,820.95	+82.6%	+82.6%	Runner-up
TCS	2,548.20 / 1,274.10*	1,941.25	-23.8%	**+52.4%**	**Strong Winner**
Infosys	1,975.80 / 987.90*	1,365.00	-30.9%	**+38.2%**	**Strong Winner**
Bajaj Auto	2,454.10	2,700.70	+10.0%	+10.0%	Moderate Gain
Hero Motocorp	3,107.30	3,293.80	+6.0%	+6.0%	Modest Gain

*Note: Pre-event start prices halved to account for 1:1 bonus share issues.

Q3: Real-World Factors Omitted from Raw CSV Data

Analysis: Raw daily price datasets omit four crucial real-world investment variables:
1. **Dividends:** Large-cap IT and auto stocks (e.g., TCS, Hero) pay significant cash dividends, boosting Total Shareholder Return (TSR).
2. **Transaction Costs & Taxes:** Brokerage commissions, Securities Transaction Tax (STT), GST, and stamp duty reduce profits on 113 executed trades.
3. **Benchmark Comparison:** Returns should be measured against the Nifty 50 or Sensex index to compute risk-adjusted alpha.
4. **Slippage & Liquidity:** Executing trades at exact close price is impossible in live markets.

Q4: Data Quality Analysis — 'Find the Holes' (Task 4)

Analysis: Task 4 queried rows with NULL `deliverable_qty`. Exactly 6 rows matched across all six tables, occurring on only two distinct dates:
• **December 9, 2015:** NULL in Eicher Motors, Hero Motocorp, TCS, and TVS Motors.
• **August 31, 2017:** NULL in Bajaj Auto and Infosys.
Conclusion: Because missing deliverable data clusters across multiple unrelated companies on identical trading days, the gap stems from exchange-level clearing/reporting failures (NSE/BSE data feed disruption) rather than missing corporate filings.

Q5: Stretch Challenge — TCS Signal Distortion Analysis

Analysis: Rebuilding TCS moving averages on adjusted prices reveals that **one original Sell signal was completely fake**:
• On unadjusted data, a **Sell signal triggered on 2018-06-05** (Close ■1,721.20) because the 1:1 bonus price drop on 2018-05-31 artificially pulled the 20-day SMA below the 50-day SMA.
• On corporate-action adjusted data, TCS maintained a unbroken **Buy signal from 2018-04-20 through July 2018**. The fake June 2018 Sell signal completely disappeared!

Part 4: SQL Verification & Checkpoint Audit

Task # & Goal	Guide Checkpoint Expectation	Our Executed SQL Result	Status
Task 1: History	889 rows, 2015-01-01 to 2018-07-31	889 days | 2015-01-01 to 2018-07-31	MATCH
Task 2: Eicher Top 5	Top close > ■32,000, all same month	Top ■32,786.40 | All 5 in Sep 2017	MATCH
Task 3: TCS Yearly	2016 avg equals exactly 2419.00	2016 avg = 2419.00 (Oldest first)	MATCH
Task 4: Find Holes	6 rows total, 2 distinct dates	6 rows | Dates: 2015-12-09, 2017-08-31	MATCH
Task 5: Moving Avg	First ma20: 2015-01-29 (2415.53)	2015-01-29 = 2415.53 | ma50=2283.80	MATCH
Task 6: Master Table	889 rows, 7 cols (2018-07-31: 2700.70/517.45)	889 rows | bajaj: 2700.70, tvs: 517.45	MATCH

Task 7: Signals	First Buy: 2015-05-18, First Sell: 2015-08-24	Buy: 2015-05-18 | Sell: 2015-08-24	MATCH
Task 8: Signal Count	3 rows (Buy, Hold, Sell), total 889	Buy: 12, Hold: 866, Sell: 11 (Sum 889)	MATCH
Task 9: Single Day	2018-06-21 signal column	2018-06-21 = Buy	MATCH
Task 10: All 6 Stocks	6 rows, 5 cols, 56 Buys, 57 Sells	6 rows | Total Buys: 56, Sells: 57	MATCH
Task 11: Returns	TVS top 86.9%, 2 stocks negative	TVS: +86.9%, TCS & Infy negative	MATCH
Task 12: Data Trap	2 stocks wildly worse than rest (-50%)	TCS -50.4%, Infosys -49.9%	MATCH
Task 13: Fix It	TCS & Infosys adjusted returns	TCS: +52.4%, Infosys: +38.2%	MATCH

Submission Readiness & Verification

✔ **SQL File:** `04_stock_market_analysis.sql` created, fully runnable top-to-bottom without errors.
✔ **PDF Report:** `Stock_Market_Analysis_Insights.pdf` created, matching all quantitative checkpoints.
✔ **Data Traps Handled:** TCS and Infosys corporate actions identified, explained, and adjusted.
✔ **Visualizations & Insights:** Every single graph paired directly with Key Insights, Recommendations, and Caveats.
✔ **No Hardcoded Placeholders:** All numbers derived directly from executed SQL queries.